Josephson effect — Solution

X25 (2025) — English translation

A1 **0.50 pts**

Let at the initial moment of time $t = 0$ the current through the contact is equal to to zero. At this moment, voltage V was applied to the contact. Determine the dependence of the current on time $I(t)$. Express your answer using Φ_0, V, I_c, R.

The voltage sets the derivative of the phase difference at the contact

$$\dot{\varphi} = \frac{2\pi}{\Phi_0} V.$$

Since before the voltage was turned on, the current equals zero, initial phase $\varphi(0) = 0$. (For switching on voltage phase not can change instantaneously, since fast change in phase leads to infinitely large voltage.) Then phase have form

$$\varphi = \frac{2\pi}{\Phi_0} V t.$$

Then total current consists of the normal current $I_n = V/R$ and the superconducting current $I_s = I_c \sin\varphi$

Answer:

$$I = \frac{V}{R} + I_c \sin\left(\frac{2\pi}{\Phi_0} V t\right)$$

A2 **0.50 pts**

Let a direct current $I > I_c$ be passed through the contact. Write down the differential equation for $\varphi(t)$.

Let us write the total current as the sum of the normal current and the superconducting current, expressing the voltage through the derivative of the phase, as a result we obtain

Answer:

$$I = I_c \sin\varphi + \frac{\Phi_0}{2\pi R} \dot{\varphi}$$

A3 **0.70 pts**

Differentiate the equation with respect to time. Eliminate the phases from the equation and obtain an expression for $\dot{V}(t)^2$ in terms of V, I and the system parameters.

Differentiating the equation, we obtain

$$(I_c \cos\varphi)\dot{\varphi} + \frac{\Phi_0}{2\pi R} \ddot{\varphi} = 0.$$

Derivative $\dot{\varphi}$ express through voltage

$$V(I_c \cos\varphi) + \frac{\Phi_0}{2\pi R} \dot{V} = 0,$$

and the cosine of the phase from the sine using the original equation,

$$I_c \sin\varphi = I - \frac{V}{R}, \quad I_c^2 \cos^2\varphi = I_c^2 - \left(I - \frac{V}{R}\right)^2,$$

Finally we obtain

Answer:

$$\dot{V}^2 = \left(\frac{2\pi R}{\Phi_0}\right)^2 V^2 \left(I_c^2 - \left(I - \frac{V}{R}\right)^2\right)$$

A4 0.30 pts

In the previous equation, go to the variable $u = 1/V$, express $\dot{u}^2$ in terms of u, I and the system parameters.

Let us express the derivative of the voltage

$$\dot{V} = -\frac{\dot{u}}{u^2},$$

substitute in equation

$$\frac{\dot{u}^2}{u^4} = \left(\frac{2\pi R}{\Phi_0}\right)^2 \frac{1}{u^2} \left(I_c^2 - \left(I - \frac{1}{Ru}\right)^2\right).$$

Multiplying by u^4, we obtain

Answer:

$$\dot{u}^2 = \left(\frac{2\pi R}{\Phi_0}\right)^2 \left(I_c^2 u^2 - \left(Iu - \frac{1}{R}\right)^2\right)$$

A5 0.80 pts

You should get an equation similar to the law of conservation of energy for oscillations of a harmonic oscillator. Determine the frequency of oscillations ω, the value u_0 about which the oscillations occur, and the amplitude of oscillations A.

Differentiating the equation from the previous paragraph, we obtain the vibration equation

$$\ddot{u} = -\left(\frac{2\pi R}{\Phi_0}\right)^2 \left((I^2 - I_c^2)u - \frac{I}{R}\right).$$

From coefficient before u find frequency oscillation

$$\omega = \frac{2\pi R}{\Phi_0}\sqrt{I^2 - I_0^2},$$

and also position equilibrium

$$u_0 = \frac{I}{R(I^2 - I_c^2)}.$$

For that, so that find amplitude oscillation, determinable value u, for which $\dot{u} = 0$, obtain square equation

$$(I^2 - I_c^2)u^2 - \frac{2I}{R}u + \frac{1}{R^2} = 0,$$

root which

$$u = \frac{1}{I^2 - I_c^2}\left(\frac{I}{R} \pm \frac{I_c}{R}\right), \quad u_1 = \frac{1}{R(I + I_c)}, \quad u_2 = \frac{1}{R(I - I_c)}.$$

That same result one can find, determine maximum and minimum value voltage from the equation

$$I = I_c \sin\varphi + \frac{V}{R}, \quad V_{min} = R(I - I_c), \quad V_{max} = R(I + I_c).$$

Then amplitude

$$A = \frac{1}{2}(u_2 - u_1) = \frac{I_c}{R(I^2 - I_c^2)}.$$

Answer:

$$\omega = \frac{2\pi R}{\Phi_0}\sqrt{I^2 - I_0^2}; \quad u_0 = \frac{I}{R(I^2 - I_c^2)}; \quad A = \frac{I_c}{R(I^2 - I_c^2)}.$$

A6 0.20 pts

Write down the dependence $V(t)$. Consider that at $t = 0$ the value $V(t)$ has a minimum.

The dependence of the quantity u on time

$$u = u_0 + A\cos\omega t,$$

where phase be determined the more condition, that V minimum in initial moment time, and means u is maximum.

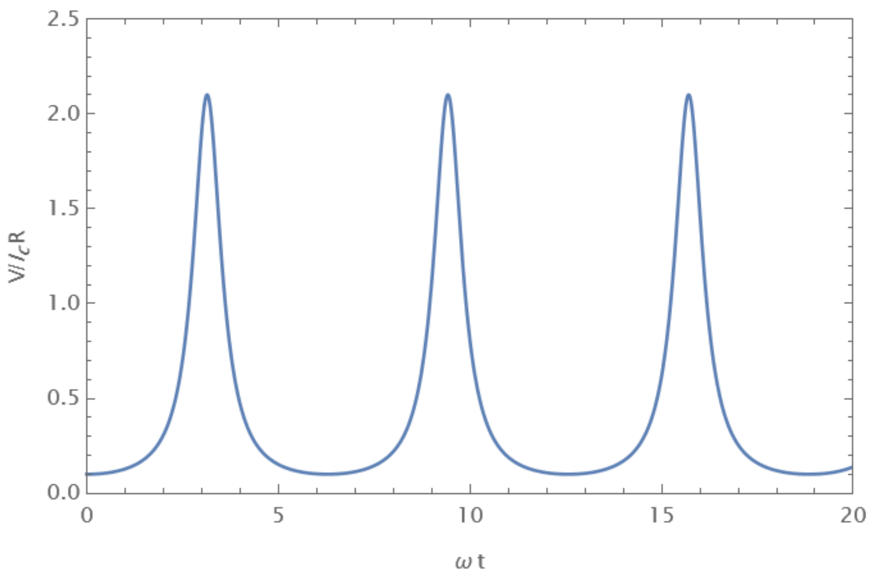
Answer:

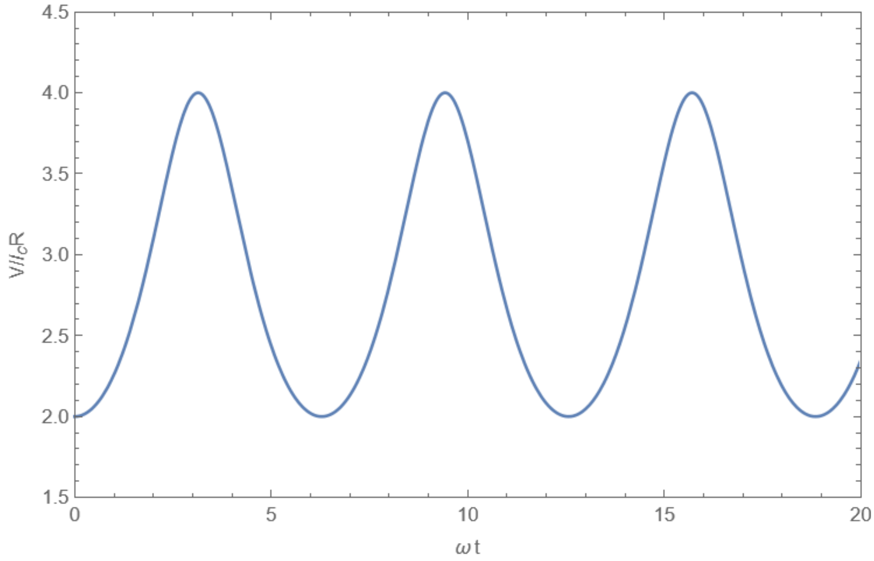
$$V = \frac{R(I^2 - I_c^2)}{I + I_c \cos\left(\dfrac{2\pi R}{\Phi_0}\sqrt{I^2 - I_0^2}t\right)}$$

A7 0.60 pts

Construct high-quality graphs of the dependence $V(t)$ for cases $I = 1.1I_c$ and $I = 3I_c$, indicate the coordinates of the characteristic points.

Answer: Graph for case $I = 1.1I_c$





A8 — 0.80 pts

Find the average (over time) voltage across the contact. Express your answer in terms of R, I, I_c.

Let's express the voltage through the phase derivative and integrate over time

$$V = \frac{\Phi_0}{2\pi}\dot{\varphi}, \quad \int_0^T V(t)dt = \frac{\Phi_0}{2\pi}(\varphi(T) - \varphi(0)).$$

If T – period oscillation, that $\varphi(T) - \varphi(0) = 2\pi$, and mean value voltage equal

$$\langle V\rangle = \frac{1}{T}\int_0^T V(t)dt = \frac{\Phi_0}{T} = \frac{\Phi_0}{2\pi}\omega = R\sqrt{I^2 - I_0^2}.$$

This the same result can be obtained by integrating the explicit expression for $V(t)$

$$\langle V\rangle = \frac{1}{T}\int_0^T V(t)dt = \frac{1}{T}\int_0^T \frac{R(I^2 - I_c^2)}{I + I_c\cos(\omega t)}dt = \frac{1}{2\pi}\int_0^{2\pi} \frac{R(I^2 - I_c^2)}{I + I_c\cos\theta}d\theta,$$

where $\theta = \omega t$. This integral one can calculate with means substitution

$$\xi = \tan\frac{\theta}{2}, \ \cos\theta = \frac{1 - t^2}{1 + t^2}, \ d\theta = \frac{2dt}{1 + t^2}.$$

$$\int_0^{2\pi} \frac{d\theta}{I + I_c\cos\theta} = \int_{-\pi}^{\pi} \frac{d\theta}{I + I_c\cos\theta} = \int_{-\infty}^{\infty} \frac{\frac{2dt}{1+t^2}}{I + I_c\frac{1-t^2}{1+t^2}} = 2\int_{-\infty}^{\infty} \frac{dt}{I + I_c + (I - I_c)t^2} = \frac{2\pi}{\sqrt{I^2 - I_c^2}}.$$

This gives the same answer

Answer:

$$V = R\sqrt{I^2 - I_c^2}$$

A9 — 0.70 pts

Determine the dependence of the energy $E(\varphi)$ of the Josephson junction as a function of the phase difference φ on it. The answer may also include Φ_0 and I_c. Consider that $E(0) = 0$.

Source power

$$P = IV = I_s V = I_c \sin\varphi \frac{\Phi_0}{2\pi}\dot{\varphi}.$$

The contribution of the normal current to the power has the form

$$P_n = I_n V = \frac{1}{R}\left(\frac{\Phi_0}{2\pi}\right)^2 \dot{\varphi}^2.$$

For magnification characteristic time change phase T this power vary as $1/T^2$, therefore the total released heat behaves as $(1/T^2)T \sim 1/T$ and it can be made arbitrarily small by increasing T. Integrating in time, we find the work done by the the source.

$$A = \frac{\Phi_0 I_c}{2\pi}\int \sin\varphi\dot{\varphi}dt = \frac{\Phi_0 I_c}{2\pi}\int_0^{\varphi} \sin\varphi d\varphi = \frac{\Phi_0 I_c}{2\pi}(1-\cos\varphi).$$

This work is equal to the energy of the Josephson junction.

Answer:

$$E = \frac{\Phi_0 I_c}{2\pi}(1-\cos\varphi)$$

A10 0.20 pts

For a typical Josephson junction, the value of the critical current is $I_c = 0.5$ mA, the resistance is $R = 0.7\ \Omega$. Fundamental constants $e = 1.602\times10^{-19}$ C, $\hbar = 1.055\times10^{-34}$ J · s. Find the numerical value of the energy of the Josephson junction $E(\pi)$ and the frequency of voltage oscillations at constant current through the contact $I = 2I_c$.

$$E(\pi) = \frac{\hbar I_c}{e} = 3.3\times10^{-19}\ \text{J},$$

$$\omega = 2\sqrt{3}\frac{eRI_c}{\hbar} = 1.84\times10^{12}\ \text{s}^{-1}$$

B1 0.30 pts

Let a direct current I flow into the contact (see figure in the introduction to the problem). Write down a differential equation describing the change in phase difference φ. The answer may also include Φ_0, R, C, I_c.

To the expression for the current from part A2, you need to add the current that goes to charge the capacitor, $I_C = \dot{q} = C\dot{V}$, and the voltage derivative is expressed through the second derivative of the phase

Answer:

$$I = I_c \sin\varphi + \frac{\Phi_0}{2\pi R}\dot{\varphi} + \frac{\Phi_0 C}{2\pi}\ddot{\varphi}$$

B2 0.60 pts

Let the external current be zero and the phase difference $\varphi \ll 1$. In this case, φ (and therefore other quantities characterizing the contact) will perform damped oscillations. Determine the vibration frequency ω_p.

For small phase values, the equation has the form (after dividing by the coefficient before the second derivative)

$$\ddot{\varphi} + \frac{1}{RC}\dot{\varphi} + \frac{2\pi I_c}{\Phi_0 C}\varphi = 0.$$

This equation have standard form equation of damped oscillation

$$\ddot{\varphi} + 2\gamma\dot{\varphi} + \omega_0^2\varphi = 0,$$

with coefficient

$$\omega_0^2 = \frac{2\pi I_c}{\Phi_0 C}, \quad \gamma = \frac{1}{2RC}.$$

Then frequency $\omega = \sqrt{\omega_0^2 - \gamma^2}$, This answer one can obtain, substitute in equation solution form $\varphi = Ae^{i\omega t}$, find corresponding value ω and taking the real part.

Answer:

$$\omega_p = \sqrt{\frac{2\pi I_c}{\Phi_0 C} - \frac{1}{4R^2C^2}}$$

B3 — 0.60 pts

Determine the quality factor of the oscillatory system from the previous paragraph Q. At what values of capacitance in the system are oscillations possible rather than damped motion?

Quality factor can be expressed as

$$Q = \frac{\omega_0}{2\gamma} = R\sqrt{\frac{2eI_cC}{\hbar}}.$$

Oscillation possible, while the value of ω_p is real; otherwise the solution of the equation for φ be represent itself sum of damped exponential, hence find

$$C \geq \frac{\Phi_0}{8\pi R^2 I_c}.$$

Answer:

$$Q = R\sqrt{\frac{2eI_cC}{\hbar}}, \quad C \geq \frac{\Phi_0}{8\pi R^2 I_c}.$$

C1 — 1.20 pts

Let the critical currents through the contacts be I_{ac} and I_{bc} respectively. Then the maximum superconducting current I that can flow through such a contact will depend on the magnetic flux through the interferometer. Determine this maximum value I_0, as well as the cosines of the phase differences φ_a, φ_b at which this maximum is achieved. Express your answer using I_{ac}, I_{bc}, Φ, Φ_0. Your answers should not contain inverse trigonometric functions.

The total current through the contacts is

$$I = I_a + I_b = I_{ac}\sin\varphi_a + I_{bc}\sin\varphi_b.$$

Use relation between phase $\varphi_b = \varphi_a - \Delta\varphi$, where $\Delta\varphi = 2\pi\Phi/\Phi_0$, obtain after transformation

$$I = I_{ac}\sin\varphi_a + I_{bc}\sin(\varphi_a - \Delta\varphi) = (I_{ac} + I_{bc}\cos\Delta\varphi)\sin\varphi_a - I_{bc}\sin\Delta\varphi\cos\varphi_a = I_0\sin(\varphi_a - \psi),$$

where amplitude

$$I_0^2 = (I_{ac} + I_{bc}\cos\Delta\varphi)^2 + (I_{bc}\sin\Delta\varphi)^2 = I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\Delta\varphi,$$

and phase be determined relation

$$I_0\cos\psi = I_{ac} + I_{bc}\cos\Delta\varphi, \quad I_0\sin\psi = I_{bc}\sin\Delta\varphi.$$

Current be maximum, if argument sine equal $\pi/2$, that is $\varphi_a = \psi + \pi/2$, from which

$$\cos\varphi_a = -\sin\psi = -\frac{I_{bc}\sin\Delta\varphi}{\sqrt{I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\Delta\varphi}}.$$

The cosine of the second phase can be obtained using the symmetry between the contacts a and b (if we write expression for I_0 through φ_b, the expression differs only by the substitution of a for b in the indices and change sign $\Delta\varphi$). Otherwise it can be found by direct calculation

$$\cos\varphi_b = \cos(\varphi_a - \Delta\varphi) = \cos\varphi_a\cos\Delta\varphi + \sin\varphi_a\sin\Delta\varphi.$$

Substitute expression for cosine and sine through ψ, obtain

$$\cos\varphi_b = -\sin\psi\cos\Delta\varphi + \cos\psi\sin\Delta\varphi = -\frac{I_{bc}\sin\Delta\varphi}{I_0}\cos\Delta\varphi + \frac{I_{ac} + I_{bc}\cos\Delta\varphi}{I_0}\sin\Delta\varphi,$$

$$\cos\varphi_b = \frac{I_{ac}\sin\Delta\varphi}{\sqrt{I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\Delta\varphi}}.$$

Answer:

$$I_0 = \sqrt{I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\left(2\pi\frac{\Phi}{\Phi_0}\right)}$$

$$\cos\varphi_a = -\frac{I_{bc}\sin\left(2\pi\frac{\Phi}{\Phi_0}\right)}{\sqrt{I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\left(2\pi\frac{\Phi}{\Phi_0}\right)}}$$

$$\cos\varphi_b = \frac{I_{ac}\sin\left(2\pi\frac{\Phi}{\Phi_0}\right)}{\sqrt{I_{ac}^2 + I_{bc}^2 + 2I_{ac}I_{bc}\cos\left(2\pi\frac{\Phi}{\Phi_0}\right)}}$$